Table 1: **All cited results.** Rows are datasets, columns are algorithms. Every cell trains on one ∼1,001,000-transition pack (the 1M-only policy: sample budget is constant by construction) for 500k gradient steps — 300k for the ManiSkill rows — with a flow actor throughout; final eval over 50 episodes. Entries are mean ± 95% confidence interval ($1.96\,s/\sqrt{n}$), n=3 seeds unless a footnote says otherwise. BC is the matched `bcflow` floor (critic q-term off); FB is vanilla Forward–Backward at the bridge's stock OGBench HPs [f,g]; biGCIQL is the bilinear critic at d=50; Density-FB is the successor-measure critic (spec 21) under the frozen soft config [d,e]. Bold marks the best mean in rows where the runner-up mean lies outside the leader's CI; "—" marks a cell that was never run. The one-line reading: the bilinear critic's only win is `cube-single`, where Density-FB matches it; on the repaired Metaworld protocol GCIQL keeps `mw-pick-place` (0.993/0.967 on the scripted resample data) while every `mw-push` cell is a statistical tie; Density-FB takes the OOD-goal split outright (0.921 vs 0.492 — coverage failure barely dents the measure) and `ms-pick-cube`, and its reference-stack numbers [d] lead `cube-double` and `antmaze-medium`. Density-FB's residual `mw-pick-place` gap is a training instability, not a ceiling (footnote e).

Dataset	BC	FB	GCIQL	biGCIQL	Density-FB
OGBench (stock 1M `play`/`navigate` data, standard task eval)					
`cube-single`	0.086 ± 0.011 [a]	0.420 [f]	0.732 ± 0.018	0.927 ± 0.019	0.915 ± 0.098 [d]
`cube-single`, OOD-goal split *	0.073 ± 0.038	—	0.492 ± 0.025	0.364 ± 0.014	$\mathbf{0.921 \pm 0.036}$
`cube-double`	0.023 ± 0.014	0.005 ± 0.016 [f]	0.377 ± 0.111	0.257 ± 0.043	$\mathbf{0.674 \pm 0.051}$ [d]
`scene`	0.048 ± 0.019	0.076 ± 0.057 [f]	0.589 ± 0.020	0.255 ± 0.020	0.641 ± 0.123 [d]
`antmaze-medium`	0.160 ± 0.024	0.593 ± 0.250 [f]	0.253 ± 0.044	0.244 ± 0.032	0.676 ± 0.188 [d]
Metaworld `play-resample` v3 (2000×500; repaired protocol †)					
`mw-pick-place`	0.440 ± 0.081	0.627 ± 0.052 [g]	$\mathbf{0.993 \pm 0.014}$	0.920 ± 0.045	0.820 ± 0.126
`mw-push`	0.540 ± 0.000	0.627 ± 0.079 [g]	0.733 ± 0.198	0.460 ± 0.294 [b]	0.753 ± 0.102
Metaworld `play-noise-hard-1m` (10000×100, same repaired protocol)					
`mw-pick-place`	0.340 ± 0.023	0.427 ± 0.114 [g]	0.520 ± 0.060	0.447 ± 0.104	0.500 ± 0.023
`mw-push`	0.327 ± 0.079	0.340 ± 0.060 [g]	0.527 ± 0.057	0.533 ± 0.069	0.420 ± 0.045
Metaworld `play-resample-hard` v3 (2000×500, `noise_max`= 0.1, same repaired protocol)					
`mw-pick-place`	0.367 ± 0.124	0.553 ± 0.035 [g]	$\mathbf{0.967 \pm 0.035}$	0.860 ± 0.141	0.673 ± 0.227 [e]
`mw-push`	0.613 ± 0.035	0.607 ± 0.111 [g]	0.813 ± 0.072	0.627 ± 0.150 [b]	0.740 ± 0.060
ManiSkill `ms-play-resample` (1M per task, masked goal channel, 300k steps)					
`ms-pick-cube`	0.020 ± 0.000 [c]	0.033 ± 0.013 [g]	0.040 ± 0.023	0.040 ± 0.000	$\mathbf{0.087 \pm 0.035}$
`ms-push-cube` (degenerate control)	0.987 ± 0.026	0.980 ± 0.023 [g]	0.600 ± 0.138	0.767 ± 0.158	1.000 ± 0.000

Notes. * OOD-goal split: training goals confined to block-$x < 0$, standard (full-space) eval — the deliberately induced goal-coverage failure. † Repaired protocol: `obs/goal_mask_dims` = [36:39] (the baked-in goal coordinates masked out of observations *and* goals), `p_randomgoal`= 0, and a *per-episode synthesized* eval goal replacing the old loader's single fixed frame, which fed every goal-conditioned critic the same wrong goal.

[a] Pooled over nine `bcflow` runs (n=9: three critic conditions × three seeds; the `bcflow` actor ignores the critic, so every such run is the same BC control).

[b] *Bimodal/wide seeds*: `mw-push` on `play-resample` v3 is 0.30/0.32/0.76 — a two-mode distribution sitting *below* its own BC floor — and on `play-resample-hard` v3 it is 0.48/0.66/0.74, the same instability in milder form; mean ± CI is a poor summary of both.

[c] The unmasked (goal-visible) BC control reaches 0.967±0.035 on `ms-pick-cube` and 0.980±0.023 on `ms-push-cube`. The masked pick-cube floor is a contact-precision limit rather than an evaluation artifact (checkpoint re-evaluation under the corrected grasp-offset synthetic goal frame reproduces 0.02–0.04, control 0.98); `ms-push-cube`'s goal is determined by the cube pose, so goal-blind BC saturates and guidance can only subtract. Neither cell discriminates architectures.

[d] *Provenance of the Density-FB OGBench cells*: pasted from the method-development (bridge) stack — reference implementation, FQL+BC flow actor, *hard* variant, same stock 1M packs, 500k steps; `cube-double` (0.700/0.648) and `antmaze-medium` (0.772/0.580) are n=2 cells. This repo's port reproduces `cube-single` at 0.916 ± 0.016, which is what certifies the port. NOTE the config seam: the four stock-OGBench rows are hard-variant reference numbers, while the OOD-split row and every row below runs this repo's frozen *soft* config (footnote e); a soft rerun of the four stock rows would unify the column. The Metaworld Density-FB cells are this repo's port under the repaired protocol (the `mw-pick-place` failure survives the pre-registered `p_randomgoal`= 0.3 ablation, 0.140 ± 0.170).

[e] *Frozen-config column (2026-08-17).* Every non-OGBench Density-FB cell (and the OOD-goal split) is a fresh

3-seed run under ONE frozen config: the *soft* variant (`measure_density_hard=false`), all else stock. This config was selected on the `play-resample mw-pick-place` dev cell after the original hard-variant column collapsed there (0.133 ± 0.057, a diagnosed actor-bootstrap training instability, not a capacity limit); soft graduated all three screen seeds (0.80/0.82/0.82, drops ≤ 0.10). Residual seed spread persists on `play-resample-hard mw-pick-place` (0.90/0.52/0.60) — the same instability in attenuated form — so mean ± CI remains a wide summary of that cell.

[f] FB OGBench cells: external provenance (reference-stack numbers, not this repo's runs); `cube-single` is a 10-seed mean reported without a CI.

[g] FB non-OGBench cells: this repo's port under the repaired protocol, rerun 2026-08-17 after fixing two FB-only harness bugs (training-side goal-channel masks were silently inert for `FBDataset`, and eval conditioned on a zero z; both fixed, all cells rerun — the pre-fix runs scored 0.000 everywhere). Caveat: FB's goal sampling is its own `train_goal_ratio` z-mix rather than the shared GCDataset goal mix, so exact goal-sampling parity with the other columns is structurally unattainable for this baseline.

How each dataset was generated.

- `cube-single` / `cube-double` / `scene` / `antmaze-medium`: stock OGBench `-play-v0` / `-navigate-v0` datasets (1000×1001 transitions), collected by OGBench's own scripted noisy plan-based oracles; downloaded, not collected here. Standard fixed `task_infos` eval.
- `cube-single` OOD-goal split: the same stock `cube-single` pack; during training, goal relabeling is restricted to goals with block-$x < 0$; eval is the standard full-space protocol.
- Metaworld `play-resample` v3: `data_collection/collect_mw_play_resample.py` — 2000 trajectories $\times$ 500 wrapper steps of task-agnostic play; the scripted Sawyer oracle chases a target re-drawn from the env's own goal space every 44 steps on a timer (never on success), matching OGBench play; `eps_max`= 0, `noise_max`= 0.02; trajectories whose puck escapes the reachable workspace are rejected and re-rolled; ∼12 relocations per trajectory, ∼24k distinct goals. Changelog in `hf_dataset/README_play.md`.
- Metaworld `play-resample-hard` v3: identical collector and relocation chaining, `noise_max`= 0.1.
- Metaworld `play-noise-hard-1m`: `data_collection/collect_mw_standalone.py` (`play-noise` regime) — 10000 episodes $\times$ 100 steps, one target per episode (no re-draws), `expert_frac`= 0, hot mixture (`eps_max`= 0.5 random-mix, `noise_max`= 0.3 Gaussian); collected 2026-08-12; ∼10k distinct goals.
- ManiSkill `ms-play-resample`: `data_collection/collect_ms_play_resample.py` — `ms-pick-cube` 8000 $\times$ 125 with in-episode target re-draws (11.0 relocations per trajectory, dwell 0.279, cube-matched structure); `ms-push-cube` 20000 $\times$ 50 with 1 relocation per trajectory, a declared structure-mismatched control (its goal is determined by the cube pose). Goal dims masked at training and eval.

Table 2: **Goal coverage of every cited dataset.** *Training goals* counts the distinct goals the data presents (for Metaworld/ManiSkill, distinct commanded targets in the observation's goal dims; OGBench never records a goal, so "—"). *In-support distance* is the operative number: for each evaluation goal, the distance to the nearest *achieved* goal in the data — the marginal hindsight relabeling actually draws from — expressed in units of that environment's own success radius, so the three suites are commensurable. Evaluation goals are the true per-episode targets under `eval_goal_mode=synth`, 50 episodes (OGBench uses its fixed `task_infos`). Every row sits at a median of 0.11 radii or below: the eval set is inside the training goal marginal on every dataset, *including all six on which the bilinear critic loses*. Condition 2 therefore holds throughout and does not explain the off-cube result; the Arm D row of Table 1 is what a genuine coverage failure looks like, and it had to be induced deliberately. Recomputed from the raw arrays by `scripts/goal_coverage_all.py`; see the map in the companion note.

Dataset	Task	Transitions	Training goals	Eval goals	In-support dist. [radii] median	p95
`play-noise-hard-1m`	`mw-push`	1,000,000	9,964	50	0.02	0.05
`play-noise-hard-1m`	`mw-pick-place`	1,000,000	10,000	50	0.04	0.16
`play-resample` v3	`mw-push`	1,000,000	23,845	50	0.03	0.21
`play-resample` v3	`mw-pick-place`	1,000,000	23,999	50	0.03	0.36
`play-resample-hard` v3	`mw-push`	1,000,000	23,849	50	0.02	0.13
`play-resample-hard` v3	`mw-pick-place`	1,000,000	23,999	50	0.03	0.31
`ms-play-resample`	`ms-push-cube`	1,000,000	39,821	50	0.01	0.02
`ms-play-resample`	`ms-pick-cube`	1,000,000	95,999	50	0.11	0.34
stock play	`cube-single`	1,001,000	–	5	0.01	0.02
stock play	`cube-double`	1,001,000	–	10	0.01	0.03
stock play	`scene`	1,001,000	–	5	0.01	0.02
stock play	`antmaze-medium`	1,001,000	–	5	0.02	0.03